DIF0109

Recap Sustitución simple
Integración por [illegible]
integración por partes
Integrales trigonométricas

Sustitución Trigonométrica

En particular: Útil cuando tenemos expresiones que involucren términos de la forma

$x^2 - a^2,\ x^2 + a^2,\ a^2 - x^2$

Nos apoyaremos en 3 identidades

$\sin^2(x) = 1 - \cos^2(x)$

$\sec^2(x) = \tan^2(x) + 1$

$\tan^2(x) = \sec^2(x) - 1$

$$\int \frac{\sqrt{16 - x^2}}{x^2}\,dx$$

$x = 4\sin(\theta)$

$dx = 4\cos(\theta)\,d\theta$

$$= \int \frac{\sqrt{16 - 16\sin^2(\theta)}}{16\cos^2(\theta)}\,4\cos(\theta)\,d\theta = \int \frac{\sqrt{16(1-\sin^2(\theta))}}{16\cos^2(\theta)}\,4\cos(\theta)\,d\theta = \int \frac{4\sqrt{1-\sin^2(\theta)}}{16\cos^2(\theta)}\,4\cos(\theta)\,d\theta$$

$$= \int \frac{\sqrt{\cos^2(\theta)}}{4\sin^2(\theta)}\,\cos(\theta)\,d\theta = \int \frac{\cos^2(\theta)}{\sin^2(\theta)}\,d\theta = \int \cot^2(\theta)\,d\theta$$

$(\tan(\theta))' = \sec^2(\theta)$

$\int \cot^2(\theta) = -\cot(\theta)$

$$= \int (\csc^2(\theta) - 1)\,d\theta = \int \csc^2(\theta)\,d\theta - \int d\theta$$

$$= -\cot(\theta) - \theta + C =$$

$$= -\frac{\sqrt{16 - x^2}}{x} - \arctan\left(\frac{x}{4}\right) + C$$



$$\int \sqrt{x^2 - 4}\,dx$$

Sustitución

$x = 2\sec(\theta)$

$dx = 2\sec(\theta)\tan(\theta)\,d\theta$

$$= \int \sqrt{4\sec^2(\theta) - 4}\cdot 2\sec(\theta)\tan(\theta)\,d\theta$$

$$= \int 2\sqrt{\sec^2\theta - 1}\,2\sec(\theta)\tan(\theta)\,d\theta$$

$$= \int 4\tan(x)\sec(\theta)\tan(\theta)\,d\theta$$

$$= 4\int \tan^2(\theta)\sec(\theta)\,d\theta$$